Rare Disease, Real Kid: The MVA Hackathon 2026

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Abstract

We developed a two part, fully automated pipeline that starts with raw FASTQ files to a ranked list of candidate germline variants of Mosaic Variegated Aneuploidy (MVA). The first part of this pipeline, called recalling workflow, runs quality control of the data, alignment to the GRCh38 reference genome, sorting, duplicate marking and germline short-variant calling using the GATK HaplotypeCaller. The second step, called VCF workflow annotates variants with Ensemble VEP, adding consequence, population frequency and clinical annotations to each variant; keeps rare protein-altering variants and ranks them using Exomiser based on the patient's phenotypes.

Running the VCF workflow on both the provided VCF and our re-called VCF showed that re-calling removed variants that were likely artifacts and kept high ranking candidate variants, demonstrating the usefulness of using the whole pipeline. This pipeline is also highly replicable as it uses only widely adopted open tools and public databases, with an estimated cost of $89 USD per genome in cloud services such as AWS and will a run-time of around 120 hours per genome.

Introduction

Mosaic Variegated Aneuploidy

Mosaic Variegated Aneuploidy (MVA) is a rare condition where fraction of an individual's cells carry aneuploidies (aberrant chromosome numbers), affecting different chromosomes at once. For patients with this condition, reported phenotypic characteristics include microcephaly, cognitive impairment, growth delays and increased tumor susceptibility.

Previous studies have reported the continuous generation of new aneuplodies in patient cells, which indicates chromosomal instability. Mutational analysis of families linked to MVA have reported biallelic germline mutations in *BUB1B*, *CEP57*, *TRIP13*, *CENATAC*, *MAD1L1*, *MAD2L1BP*, *CEP192*. All of these genes are involved in the spindle-assembly checkpoint or in the chromosome segregation process during mitosis.

Approach description

Rationale

MVA is reported and usually inherited in a recessive manner where both copies of the causal gene are affected. We therefore looked for homozygous or compound heterozygous (two different variants each affecting one copy of the gene) variants that affected in a deleterious manner a protein encoded by a gene related to chromosome segregation.

Identification of germline genetic variants associated with mosaic variegated aneuploidy

In order to identify germline genetic variants that could potentially explain the phenotypes of the patient we designed and ran a bioinformatic pipeline comprised of two workflows. These two workflows match the two different inputs provided for this challenge: FASTQ files and a VCF file.

The first workflow, called recalling workflow, works with FASTQ files as input and outputs a germline VCF file. The second workflow, called VCF workflow, starts with a VCF file with germline variants and performs variant filtering, annotation, and variant priorization to output ranked rare pathogenic variants that may be associated with the phenotype of interest.

The VCF workflow was run with both the provided dataset's VCF and the VCF obtained from the recalling workflow.

The recalling workflow performs a quality control analysis of the FASTQ files using the tool FastQC in order to identify if further processing of the data is needed, such as trimming of low quality sequences or removal of adaptors from the sequencing process. After FASTQ QC analysis, alignment to the GRCh38 reference genome was performed using bwa mem, which generates BAM files. This files then undergo post-processing which include marking of PCR duplicates and sorting using the tools samtools and picard's MarkDuplicates. A post-processing QC is also performed on the output BAMs to identify any issues that could affect variant calling such as: high levels of PCR duplicates, high levels of reads with mapping quality of 0 or high levels of unmapped reads.

Afterwards, germline short variants are called using the tool HaplotypeCaller from the gatk following tool's recommendations for parameter selection. Variant filtering is then perfomed based on the tool's recommendation and using public references of known snps and indels, keeping only variants that pass the tool's filters and variant that are at sites with a minimum coverage of 10 reads. Figure 1 shows a flow diagram that summarizes this workflow.

After this step the outputted VCF is used as input in the VCF workflow.

After quality filtering, variants are annotated for consequence prediction, population frequency and clinical significance using Ensembl Variant Effect Predictor (VEP). For pathogenicity prediction scores from SIFT and Popyphen included in VEP are included

in the annotation. For population frequency public data from gnomad v3.1.2 was used. For clinical significance data from ClinVar is used.

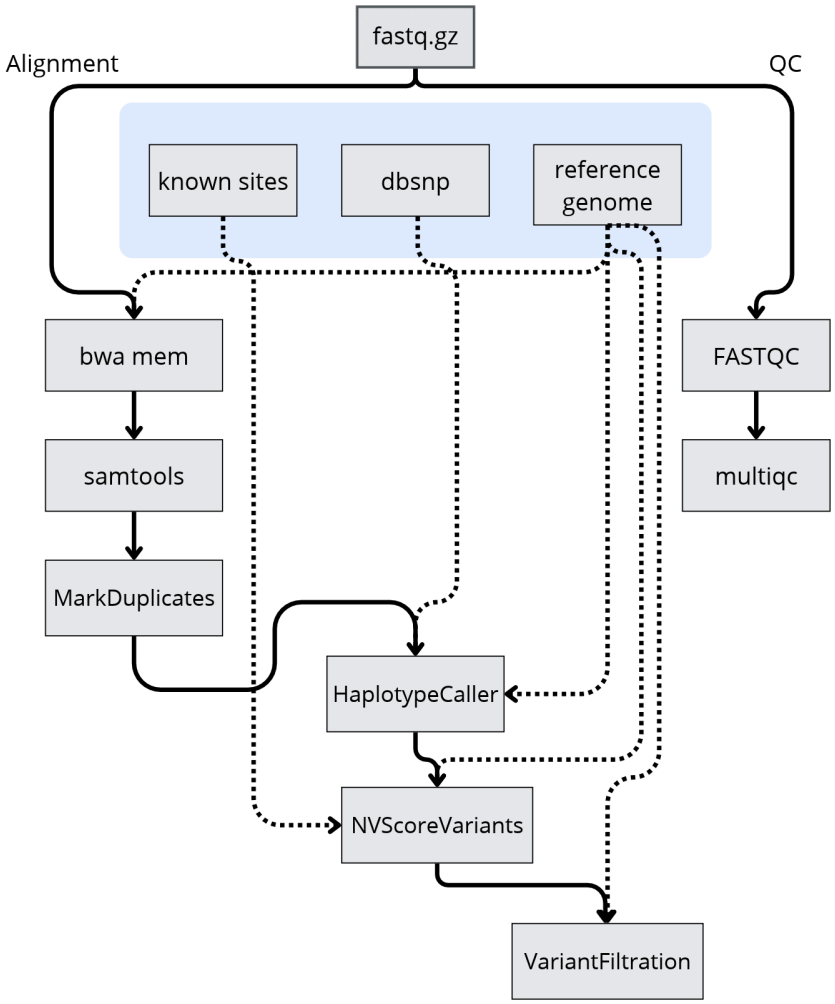


Figure 1. Recalling workflow

As the phenotype of interest is a rare condition, the pipeline then filters the VCF to keep variants that are reported with low frequency (AF = 0.001, 0.1%) in the gnomad database or that do not have any annotation of population frequency.

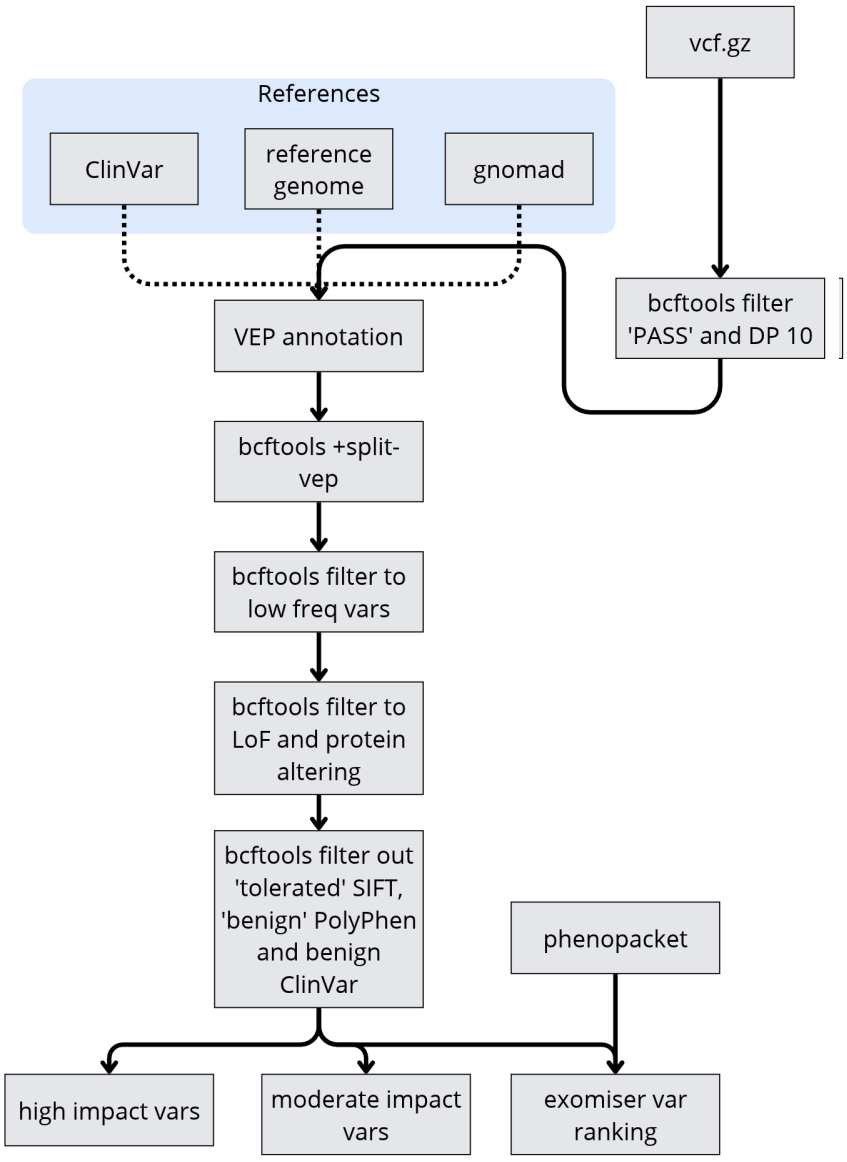


Figure 2. VCF workflow.

Next, the pipeline will filter the VCF to keep variants that have predicted consequences of loss of function or protein alteration (such as stop gains, frameshift variants, missense variants and inframe deletions or insertions).

The VCF will be the filtered further taking out variants that were predicted to have a tolerated or benign change on the protein. For this filtering the predictions of SIFT and PolyPhen are used. This tools predict the effect of missense variants on the coded protein. SIFT predicts if an amino acid change will affect the protein function based on the sequence homology and physical and chemical similarity between the amino acids,

while PolyPhen predicts the effect using several annotations regarding protein structure. As a last filter, the workflow eliminates variants that are annotated as benign or likely benign by Clinvar.

This filtering process was created based on the hypothesis that the phenotype of interest is being caused by the loss of function of a gene related to chromosome segregation keeping variants of high and moderate impact.

To prioritize potential causal variant candidates the workflow uses the tool exomiser. This tool uses the filtered VCF and a set of phenotypes as input and ranks variants based on the predicted pathogenicity, population frequency and if it matches to known phenotypes from human and model organism data. This prioritization is done by gene under each inheritance mode, therefore also reporting compound heterozygous pairs. Using the output of exomiser the workflow processes the data to generate a csv file with candidate variants for the phenotype of interest and adds variants that although may be unrelated to the phenotype of interest they are associated with other pathologies or are predicted to be pathogenic. Figure 2 shows a flow diagram summarizing the VCF workflow.

Choosing between the provided and the re-called VCF

For this challenge we inputed both the VCF obtained from the recalling workflow and the VCF already provided into the VCF workflow and compared results manually. Although results were similar between both cases, we did notice some variants from the VCF provided that passed all filters that failed in the gene *SERPINA1*. After careful inspection of the variants it seemed that some of this variants although reported separately where actually a mnv and therefore their consequence may have nor been estimated correctly. We also noticed that this variants were annotated as heterozygous in the VCF although some of them had a low proportion of reads supporting the alternate variant (for example 10 reads supporting the alternate allele vs 42 supporting the reference allele), this could be due to difference in the parameters used in the variant calling process. As we also saw that the mapping quality of reads supporting the alternate allele was lower than the ones supporting the reference allele we considered them as potential artifacts that were filtered in the VCF obtained from the recalling workflow. With this information in consideration we decided to use the VCF or the recalled workflow to generate final results as it demonstrated to filter better this type of artifacts.

The final set of proposed variants was therefore generated by using both workflows starting from the FASTQ files.

Automated or manually curated output

The submission file was obtained automatically by the two workflows with no manual editing of candidate variants. The only manual step was the comparison between

results obtained from the VCF generated by the recalling workflow and those form the VCF provided.

Public data used

This approach used only public available data which included:

- gnomAD v3.1.2 - population allele frequencies
- GATK resource bundle: known common SNP and indel VCFs for GRCh38
- Ensembl Variant Effect Predictor (VEP) cache and bundled databases
- ClinVar
- Exomiser data release

Capacity of outputting proposed pairs of compound heterozygous candidate variants

This approach is able to output proposed pairs of compound heterozygous candidate variants as prioritization using Exomiser is done based on genes and not only on individual variants, so two heterozygous variants in the same gene are reported together. Exomiser output was then automatically used to create the csv file for submission.

This workflow was not able to confirm if this compound heterozygous variants were in trans (one on each copy of the gene) through phasing. No parental data were available, and the variants were too far apart to be phased from the same sequencing reads.
As this is an important step to associate this variants to a loss of function of both copies of the gene, a phasing workflow is included in the GitHub repository for cases where variants that are closer to each other.

Secondary or incidental findings

We define secondary findings as rare germline variants unrelated to the MVA phenotype that are either associated with another disease or predicted to have a high-impact effect on the protein. Homozygous, heterozygous and compound heterozygous variants were all considered.
These variants were also prioritized using the tool Exomiser.

Run time and cost estimates

The whole pipeline (both workflows) has an estimated 120h wall-clock time with an estimate of cloud billing of $89.09 USD considering whole VM rental (AWS m6i.4xlarge) without considering storage costs.

Generative AI used

Anthropic, Claude Oppus, commercial terms, no training on customer content. Used to help write the code for the pipeline and debug.

Strengths of the approach

This approach generated two workflows that allow starting the analysis from different input files. Both workflows use standard and comparable tools, so results are easier to reproduce. Both workflows were ran outside from an HPC using a workstation with 12 cores and 62 GB of RAM, showing a modest need of hardware. The approach reports candidate variant pairs automatically.

Limitations of the approach

Running the workflows requires enough storage space to keep references and data from the public databases used for annotation requiring al least 3TB of space. Phasing the variants of interest was not possible as we did not have sequencing data from the parents and variants were not close enough to estimate if the could be on the same haplotype. The workflows are only designed to identify and rank short germline variants, therefore germline copy number changes and germline structural variants associated to the condition will be missed. This approach is for research only as it candidate variants need to be further studied and validated, the approach is not meant to be for clinical use.

References:

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